Nicholas Reardon

Gameplay and Systems Programmer

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Summary

Gameplay and systems programmer building production-style Unreal Engine 5 systems on Lyra, focused on C++, modular data-driven architecture, combat, encounters, persistence, and designer-facing workflows.

Skills

Languages: C++, C#, Python, GDScript, Java, SQL

Engines and Frameworks: Unreal Engine 5, Unity, Godot, Qt

Unreal Engine: Gameplay Ability System, Game Feature Plugins, Lyra, Enhanced Input, Behavior Trees, EQS, UMG, CommonUI, World Partition, MetaSound, Chaos Physics, Blueprint

Gameplay and Systems: Data-driven Design, Modular Combat, Encounter Systems, Spawn Systems, Save and Persistence, AI State Machines, Authoring Tools

Tooling: Git, Diversion, JetBrains Rider, Visual Studio, UnrealVS, Cloudflare, Doxygen, Engine Migration, Debugging, Profiling

AI and Decision Systems: Behavior Trees, EQS, Reusable Behavior Subtrees, AI Perception, A* and BFS Pathfinding, Decision Trees, scikit-learn

Graphics: Shader Fundamentals, GLSL, PBR Concepts, Day-Night and Sky Systems

3D Asset Pipeline: Blender, Modeling, UVs, Texturing, Rigging, Animation

Projects

Tethered — Project page Sept 2025 – present

- Architected a modular Lyra-based gameplay framework using Game Feature plugin isolation to keep combat, game modes, and content decoupled while iterating on an arena-mode vertical slice.
- Designed a world-level event bus on the GameState Ability System Component to coordinate encounter, spawn, and travel systems at runtime without coupling individual components.
- Built a seamless-travel subsystem that snapshots and reconstructs player and world state across level transitions, preserving GAS attributes, equipment, and run upgrades through GameState teardown.
- Implemented a pluggable save backend behind an interface, isolating disk I/O from runtime state and leaving a clean extension point for a future server-backed implementation.
- Designed a designer-facing combo system using an input-tag routing map where each step resolves attacks from owned Gameplay Tags at runtime, letting upgrades modify movesets without branching logic.
- Built data-driven encounter, wave-advance, spawn-modifier, and combo-routing systems so new encounter rules, enemy behaviors, and upgrades can be authored through data.
- Migrated a GAS-based Lyra project across two major Unreal Engine versions (5.6 to 5.8), resolving API deprecations, build and registration breakage, and Lyra base-class changes.

Alien Survivors — Project page | Play on itch.io Sept 2024 – Dec 2024

- Built an enemy spawning director with spawn pacing, cap management, and optional object pooling for performance under high enemy counts.
- Implemented rarity-weighted enemy selection with per-type parameters and runtime tuning hooks for spawn rate and movement speed.
- Built a fully data-driven weapon and upgrade framework with stat scaling and functional modifiers authored as editor-defined upgrade definitions.
- Created dynamic runtime UI that adapts to available upgrades and player state as new weapons and modifiers are added.

Last Oasis — Project page | Play on itch.io 2025 game jam

- Built a nested state-machine character controller with ground, air, wall-slide, slam, destroy, and death states; coyote time; variable-height jumping; wall jumps; input buffering; spike grace frames; and tile-data-driven slippery surfaces.
- Built a chain-reaction destructible system where a facing raycast breaks a target and an expanding overlap query breaks neighbors outward with per-depth delay.
- Created an editor-time tile-to-scene tool that swaps tiles for scene instances keyed by tile custom data while preserving flip, transpose, and rotation values.
- Built an aimable launch ability with aim, fire, and cancel states, a live guide line, and on-screen control prompts.

A Totally Normal Bike Ride — Project page | Play on itch.io 3-day game jam

- Built a two-body physics character using a wheel and rider rigidbody joined by a pin joint, with torque-based locomotion and separate ground and air balance tuning.
- Implemented a charge-and-release variable jump with compression, impulse scaling, contact-based ground state, and rolling average-speed feedback.
- Built a telephone-wire authoring tool that creates segmented physics ropes between anchors and renders them as smooth baked Curve2D lines.
- Implemented a shader-driven day-night cycle with four-phase sky blending, scripted sun and moon motion, stars, and tinted clouds.

Stock Backtesting Course project

- Built a PySide6 desktop interface for selecting ETFs, date ranges, and strategies, with embedded Matplotlib charts for price history and strategy performance.
- Designed a data-access abstraction with swappable adapters for Yahoo Finance, local JSON, Alpha Vantage, and Polygon-style sources, so adding a provider does not touch consumer code.
- Added dynamic strategy loading via Python module discovery so a new strategy drops in as a file with no controller changes, and ran data downloads on a background QThread to keep the GUI responsive.
- Implemented SMA Crossover, MACD, and Bollinger Bands strategies on shared infrastructure for tracking cash and shares, logging trades, computing return metrics, and writing CSV outputs.

Education

California State University, Fullerton, Bachelor of Science in Computer Science – Fullerton, CA Expected Fall 2026

Saddleback College, Associate of Science in Computer Science – Mission Viejo, CA May 2024

Experience

Electronics Support Technician, Cheapest iPhone Repair – Laguna Niguel and Irvine, CA June 2023 – July 2024

- Independently diagnosed and repaired unfamiliar devices through structured troubleshooting, fault isolation, and verification testing.
- Executed delicate device teardowns with detailed tracking and documentation for reliable reassembly.
- Built internal repair documentation and improved workflows to increase consistency and efficiency.